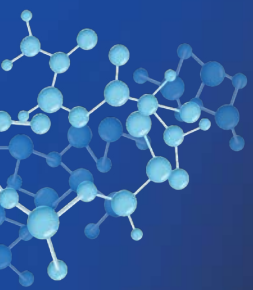
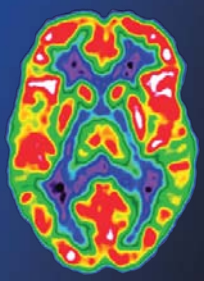
Impact on the body





BRAIN

High levels of cortisol can harm the brain by disrupting synapse regulation, resulting in the loss of sociability. Some studies have found that stress can kill brain cells and even reduce the size of the brain. Stress-induced cortisol surges in the body were found to impair memory retrieval as stress has a shrinking effect on the prefrontal cortex

HEART

Inflammation produced by high levels of cortisol in the body causes damage to the coronary arteries. Long-term exposure to cortisol also raises blood pressure, cholesterol and triglycerides. A 2019 study in *BMJ* concluded stress to be associated with early onset of cardiovascular disease

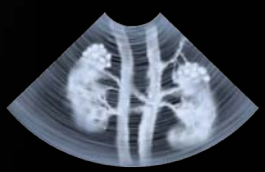




DIABETES

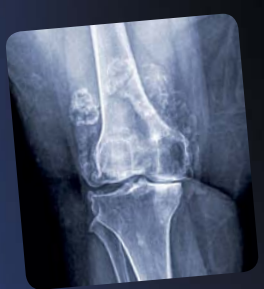


A study in the journal *BMC* in 2018 noted that 20–40% of those with Type 2 diabetes have experienced stressful events in their life. In 2020, a study in *Psychoneuroendocrinology* confirmed that cortisol increased blood sugar levels in those with Type 2 diabetes



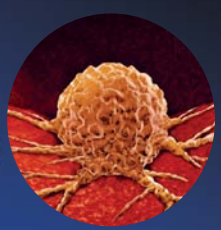
KIDNEY

Though the impact is not direct, stress leads to diabetes, which, in turn, causes chronic kidney disease in the long term



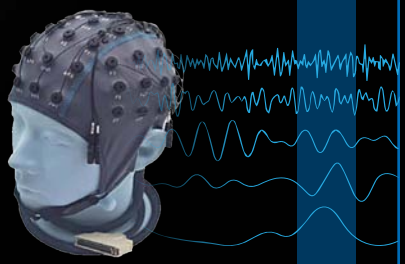
MUSCLES AND BONES

Cortisol has been shown to directly reduce muscle mass and increase risk of osteoporosis (loss of bone density). This can exert pressure on the spine, knees and lead to osteoarthritis. Stress can also cause inflammation in joints leading to joint and muscle pain



CANCER

A 2021 study in Pubmed found that stress hormones could awaken dormant cancer cells in experimental mice, thereby raising the possibility that stress could lead to cancer recurrence in humans



INSOMNIA

The first and most distinct consequence of daily mild stress is an increase in rapid-eye-movement (REM) sleep, which causes intense brain activity. Stress causes less of non-REM or deep sleep, crucial for the brain to rest and restore. Stressed people also find it harder to fall asleep due to increased cortisol in the body which doesn't let the brain relax



HORMONAL HEALTH

Stress has been directly linked to lower infertility and sex drive as cortisol inhibits the production of sex hormones. It also inhibits production of the luteinizing hormone, which is crucial to regulate women's menstrual cycles

NUTRITION

The brain reward centre starts to associate food intake with lowered stress response. Stress eating is a common condition, which results in poor nutritional intake, erratic eating patterns and often overeating



1 2 3 4 5 6 7 8 9